Lucas A. Saavedra

Contact and Personal Information

Full name Lucas Ariel Saavedra

Date of birth (dd/mm/yyyy) 30-04-1999 (27 years old)

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Education

2018 – 2023 **Computer Engineering**, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina.
Final work: *Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories.*
GPA: *8.02/10.*

Research experience

January 2022 – To date **Research Assistant.** Molecular Neurobiology Laboratory, BIOMED UCA-CONICET, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina.

- Laboratory head: Prof. Francisco J. Barrantes.
- Implementation of Deep Learning models for the analysis of static and dynamic properties of the receptor for the neurotransmitter acetylcholine (nAChR) with orientation to graph theory, artificial neural networks and geometric computing.
- Analysis of distribution and density of nicotinic acetylcholine receptors (nAChRs) in the plasma membrane of CHO-K1/A5 cells using super-resolution microscopy (STED and STORM). Advanced analytical methods implemented with MATLAB, based on graph theory and computational geometry to study the topography of nanoclusters.
- Development of an efficient and autonomous algorithm based on binary classifications with supervised Graph Neural Networks (GNNs) to detect and quantify the dynamic clustering of particles in membrane proteins in single-molecule microscopy (SMLM) data. The algorithm was developed with TensorFlow.

Research experience (continued)

- Analysis of sub-millisecond, two-colour MINFLUX single-molecule tracking data of nicotinic acetylcholine receptors and cholesterol in live cells. Segmented trajectories into confined and free-diffusion states, classified diffusive regimes (subdiffusive, Brownian, superdiffusive) via time-averaged MSD fitting adapted to irregular sampling intervals, and applied deep learning to characterise confinement.
- Developed WadTCN, a deep learning architecture for pointwise and trajectory-level analysis of 2D single-particle diffusion: anomalous exponent and diffusion coefficient estimation, model classification and trajectory segmentation. It was applied to nAChR trajectories in live cells.
- Application of AlphaFold3-equivalent models to study protein-ligand structures (specifically, the nAChR and cholesterol), combined with coarse-grained and all-atom molecular dynamics (MD) simulations, using CHARMM-GUI to prepare the systems, the GROMACS software, and high-performance computing (HPC) platforms with SLURM. Implemented deep learning methods to analyse the receptor's affinity for cholesterol, and used PyLipID to characterise cholesterol binding sites in the simulations.
- Design and programming in C++ of electronic circuits with Arduino to control piezo motors.

January 2025 – January 2026

Research Assistant. Department of Data Science, Data Science and AI Laboratory, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina.

- Supervisor: Prof. Antonio Quintero-Rincón.
- Analysis of electroencephalogram (EEG) data during mental arithmetic tasks using self-supervised learning (specifically, hierarchical variational autoencoders) to detect specific cognitive activity with PyTorch and MNE-Python.

Research Publications

Full-Length Publications in International Peer Reviewed Journals

1. **L. A. Saavedra** and F. J. Barrantes, "Machine learning in single-molecule tracking analysis of superresolution optical microscopy data," *Cells*, vol. 15, no. 8, p. 686, 2026, ISSN: 2073-4409. DOI: 10.3390/cells15080686
2. **L. A. Saavedra**, A. Quintero-Rincón, and H. Batatia, "A self-supervised reconstruction framework for segment-level detection of task-induced deviations in multi-channel eeg," *PLOS ONE*, vol. 21, no. 10, e0361586, 2026. DOI: 10.1371/journal.pone.0361586
3. G. Muñoz-Gil et al., "Quantitative evaluation of methods to analyze motion changes in single-particle experiments," *Nature Communications*, vol. 16, no. 1, p. 6749, 2025, ISSN: 2041-1723 (Electronic) 2041-1723 (Linking). DOI: 10.1038/s41467-025-61949-x
4. F. Reina, **L. A. Saavedra**, C. Eggeling, and F. J. Barrantes, "Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond minflux-based single-molecule tracking," *Nature Communications*, vol. 16, no. 1, p. 6336, 2025. DOI: 10.1038/s41467-025-61489-4
5. **L. A. Saavedra** and F. J. Barrantes, "Temporal convolutional networks work as general feature extractors for single-particle diffusion analysis," *Journal of Physics: Photonics*, vol. 7, no. 2, p. 025 017, 2025, ISSN: 2515-7647. DOI: 10.1088/2515-7647/adbec8
6. **L. A. Saavedra**, A. Mosqueira, and F. J. Barrantes, "A supervised graph-based deep learning algorithm to detect and quantify clustered particles," *Nanoscale*, vol. 16, no. 32, pp. 15 308–15 318, 2024, ISSN: 2040-3364. DOI: 10.1039/D4NR01944J

7. **L. A. Saavedra**, H. Buena-Maizón, and F. J. Barrantes, "Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies," *International Journal of Molecular Sciences*, vol. 23, no. 18, p. 22, 2022. DOI: 10.3390/ijms231810435

Presentations at International Conferences

1. M. F. Colavitta, P. G. Sanz, **L. A. Saavedra**, L. Grasso, and F. J. Barrantes, "Environmental enrichment delays the onset of cognitive decline in a transgenic mouse model of alzheimer disease," in *XL Annual Meeting SAN*, 2025.
2. **L. A. Saavedra** and F. J. Barrantes, "Machine learning empowers static and dynamics clustering characterization of transmembrane receptors," in *Building Bridges in Computational Biophysics (BBCB) v3.0*, 2024.
3. F. J. Barrantes, A. Mosqueira, H. Buena-Maizón, and **L. A. Saavedra**, "The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor," in *11th International Weber Symposium*, Punta del Este, Uruguay, 2023.

Media Coverage and Appearances

2025
- **Hagamos Lío** (Streaming, program #299, in Spanish) — Neuroscience segment, September 22. youtu.be/BKwfrr4AbMU
- **Infobae** (newspaper, in Spanish) — *"Por primera vez, científicos argentinos visualizaron en tiempo real un receptor clave vinculado al Alzheimer"*, September 10. infobae.com
- **Clarín** (newspaper, in Spanish) — *"Con inteligencia artificial, científicos argentinos hicieron un importante hallazgo para el Alzheimer"*, September 10. clarin.com

Work experience

May 2025 – To date — **Senior Full-Stack AI Developer.** Accenture, Buenos Aires, Argentina.

- LLM-based multi-agent systems (LangGraph, FastAPI, Azure OpenAI, Elasticsearch) for automated code migration between languages, data-quality testing, and AI-assisted accounting reconciliation.
- Dynamic social-network analysis platform combining a Neo4j knowledge graph and LLMs, including community detection of messages and users. Spring Boot batch classification pipeline and a LangGraph conversational agent exposed through a Next.js chat interface, plus an MCP (Model Context Protocol) server exposing the Neo4j graph as agent-consumable tools.
- Real-time computer vision systems (YOLO, OpenCV, TensorFlow) for vehicle detection, customer profiling, and dataset preparation for gas-station clients.
- Full-stack development with Next.js/React and enterprise Angular applications, deployed with Docker and Azure Pipelines.
- Research on Small Language Models (SLMs).
- Development of agents for automating the Software Development Life Cycle (SDLC) through Spec-Driven Development.
- CCTV camera-movement detection with computer vision: siamese neural networks (TensorFlow) comparing the current scene against a reference to detect angle changes, defocus, or scene changes while tolerating minor variations.

Work experience (continued)

December 2023 – May 2025 — **Freelancer.** UpWork (Click here to see my profile).

- Web scraping scripts with Python and Selenium, hosted on AWS Linux EC2 instances.
- YOLO training to identify basketball players in match videos from a European league, with a SQL database on Supabase fed by a handcrafted scraper of the FIBA website.
- Napari plugins (GUIs) combined with scikit-learn decision-tree classifiers to extract information from laser-spot images provided by the client.

June 2021 – June 2022 — **Back-End Developer.** Deloitte, Buenos Aires, Argentina.

- Threat Intelligence back-end developer, working under Agile methodologies (Scrum).
- Developed Codex Gigas (a malware "DNA" profiling search engine) and ImageSimilarity (an SSIM-based image-grouping tool), plus ThreatConnect apps.
- Built with Python, Docker, MongoDB, and SQL.

January 2021 – June 2021 — **Cyber Risk Advisory Intern.** Deloitte, Buenos Aires, Argentina.

- Offensive/defensive security and threat-intelligence fundamentals.

Teaching experience

March 2024 – June 2024 — **Assistant professor.** Discrete mathematics, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina. Main topics: Number theory, Graph theory, Demonstration methods, Boolean algebra.

July 2023 – December 2024 — **Assistant professor.** Data oriented programming, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina. Main topics: object-oriented programming (OOP), NumPy, Pandas, Python, Java, SQL, software design with UML.

March 2023 – June 2025 — **Assistant professor.** Introduction to Data Science, Pontifical Catholic University Argentina (UCA), Buenos Aires, Argentina. Main topics: definitions and concepts of artificial intelligence, introduction to artificial neural networks, scientific method.

Certificates and Courses

Certifications

2025 — **IELTS Score: 7.5**

Duolingo English Test (DET) Score: 135 (Advanced: CEFR C1)

2022 — **AWS Certified Cloud Practitioner**

Courses

2026 — **Golang for Beginners**, Udemy

Kubernetes for the Absolute Beginners - Hands-on, Udemy

2025 — **Basic to Advanced: Retrieval-Augmented Generation (RAG)**, Udemy

Bioinformatics; Learn Docking & Mol Dynamics Simulation, Udemy

Introduction to Statistics, Stanford Online, Coursera

2024 — **Deep Neural Networks with PyTorch**, IBM, Coursera

2022 — **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera

DeepLearning.AI Tensorflow Developer, DeepLearning.AI, Coursera

Certificates and Courses (continued)

- **Data Science**, Coderhouse

2021

- **Machine Learning**, Stanford Online, Coursera
- **Programming Foundations: Design Patterns (2013)**, Linkedin Learning
- **R Programming**, Johns Hopkins University, Coursera
- **Docker for the Absolute Beginner - Hands On - DevOps**, Udemy
- **Programming Foundations: Design Patterns (2013)**, LinkedIn Learning

2020

- **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera
- **Neural Networks and Deep Learning**, DeepLearning.AI, Coursera
- **CS50's Introduction to Artificial Intelligence with Python**, Harvard, edX